8.033 Fall 2025: Solution Set 2

1

a. We can use the definition of natural units, given in your lecture

$$\begin{aligned} l_p &= 1.6 \cdot 10^{-35} m \\ m_p &= 2.2 \cdot 10^{-8} kg \\ t_p &= 5.4 \cdot 10^{-44} s \end{aligned} \tag{1}$$

to convert SI units to natural units. Namely,

$$\begin{aligned} 1m &= 6.3 \cdot 10^{34} l_p \\ 1kg &= 4.6 \cdot 10^{7} m_p \\ 1s &= 1.9 \cdot 10^{43} t_p. \end{aligned} \tag{2}$$

Then we get

$$5J = 5kg \cdot m^2 \cdot s^{-2} = 5 \times 4.6 \cdot 10^7 m_p \times [6.3 \cdot 10^{34} l_p/(1.9 \cdot 10^{43} t_p)]^2 = 2.5 \cdot 10^{-9} m_p \cdot l_p^2 \cdot t_p^{-2}, \tag{3}$$

and similarly

$$10kg \cdot m^{-3} = 1.8 \cdot 10^{-96} m_p \cdot l_p^{-3}, \tag{4}$$

and

$$3m \cdot s^{-2} = 5.2 \cdot 10^{-52} l_p \cdot t_p^{-2}. \tag{5}$$

b. Using Eq. (1), we can convert natural units to SI and get

$$\begin{aligned} 5l_p \cdot t_p^{-1} &= 1.5 \cdot 10^9 m s^{-1} \\ 3t_p &= 1.6 \cdot 10^{-43} s. \end{aligned} \tag{6}$$

2

The Lorentz transformations, with factors of c inserted, are

$$x' = \gamma(v)(x - vt) = \left[1 - \left(\frac{v}{c}\right)^2\right]^{-\frac{1}{2}} (x - vt) \tag{7}$$

$$t' = \gamma(v)\left(t - \frac{v}{c^2}x\right) = \left[1 - \left(\frac{v}{c}\right)^2\right]^{-\frac{1}{2}} \left(t - \frac{v}{c^2}x\right). \tag{8}$$

Using the formula $(1+\epsilon)^\alpha = 1 + \alpha\epsilon + \mathcal{O}(\epsilon^2)$ where $\epsilon \ll 1$, we can expand Eq. (7) and find

$$x' = \left[1 + \frac{1}{2}\left(\frac{v}{c}\right)^2 + \mathcal{O}\left(\left(\frac{v}{c}\right)^4\right)\right](x - vt). \tag{9}$$

Dropping all the terms that involves (v/c) since we are considering the case where $v \ll c$, we get

$$x' \approx x - vt. \tag{10}$$

Similarly, we expand Eq. (8) and find

$$t' = \left[1 + \frac{1}{2}\left(\frac{v}{c}\right)^2 + \mathcal{O}\left(\left(\frac{v}{c}\right)^4\right)\right]\left(t - \frac{v}{c^2}x\right). \tag{11}$$

Again, dropping terms that involves (v/c), we get

$$t' \approx t. \tag{12}$$

Equations (10) and (12) are indeed Galilean transformations.

3

a. The order in which an observer sitting at rest at the spatial origin received light from the flashlights can be determined from the order in which the worldlines of light reaches the t-axis in Fig. (1). The order from earlier to later is A, C, D=B, and E.

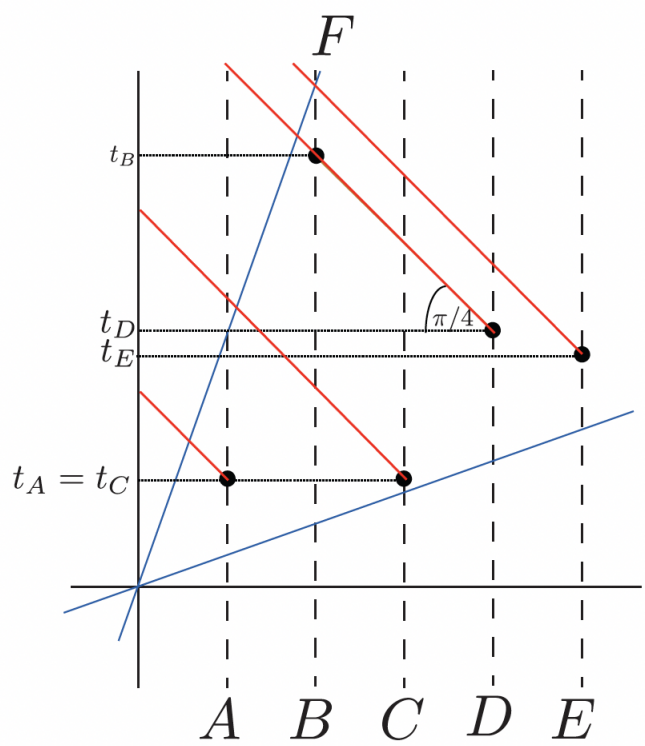

Figure 1: Alice, Bob, Charlie, Diana, and Eli turn on their flashlights. The red lines indicate the worldlines of light from their flashlights.

b. The order in which the flashlights turn on in Fred's frame can be determined by adding lines of simultaneity (lines parallel to Fred's x-axis) through the events in which Alice, Bob, Charlie, Diana, and Eli turn on their flashlights (see Fig. (2)). The order from earlier to later, measured in Fred's frame (which we can tell by the order of the lines of simultaneity intersecting Fred's t-axis), is C, A = E, D, and B.

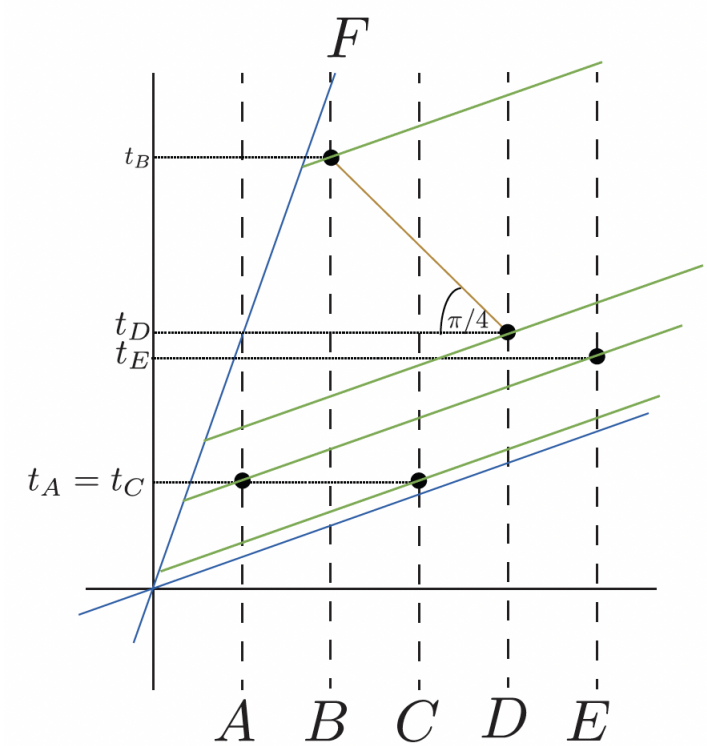

Figure 2: The green lines are the lines of simultaneity in Fred's frame.

c. The worldlines of light in Fred's frame are the same as shown in Fig. (1). We can tell the order in which light reaches Fred by comparing the order in which the worldlines of light intersects Fred's t-axis in Fig. (1). The order from earlier to later is A, C, D=B, and E, which is identical to the answer to part a. This is a consequence of the invariance of the speed of light for inertial observers.

4

It is rather straightforward to use a spacetime diagram (see Fig. (3)) to show that neither you or your friend agree with the referee. In the referee's frame, you crossing the finish line (event A) and your friend crossing the finish line (event B) occurs at the same time (notice that the referee's line of simultaneity goes through both event A and B). However, your line of simultaneity going through event A intersects your friend's worldline at event B', in which your friend is still on her way to the finish line. Therefore, you observe that your friend has not reached the finishing line yet when you have, so you conclude that you have won the race. The situation is similar in your friend's frame. The line of simultaneity in your friend's

frame going through event B intersects your world line at event A', in which you have not reached the finish line yet, so your friend thinks that she has won the race.

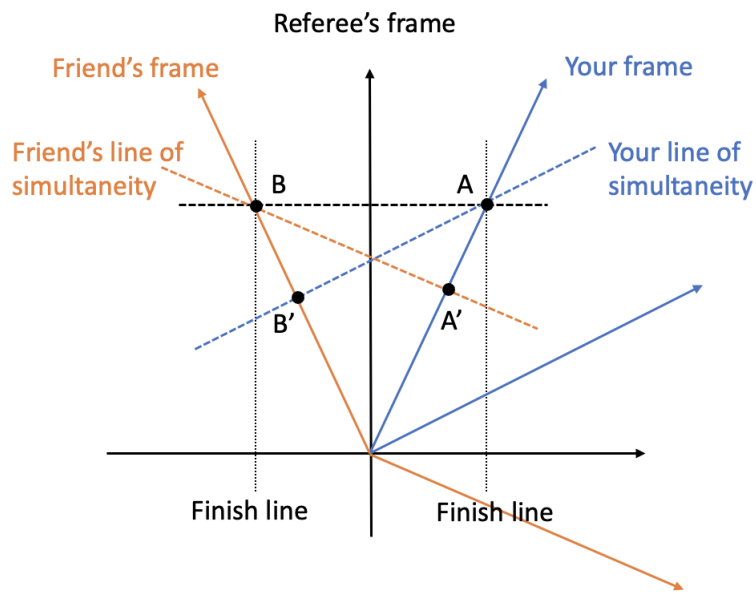

Figure 3: The spacetime diagram that illustrates problem 4. The black, orange, and blue axes indicate the referee's, your friend's, and your frame, respectively. Similarly, the dashed lines in black, orange, and blue correspond to the line of simultaneity in the referee's, your friend's, and your frame, respectively. You and your friend started the race at the origin and each crosses the finish line at event A and B, respectively. The lines of simultaneity in your and your friend's frame show that you and your friend do not agree on who wins the race.

5

Similar to the previous problem, it is straightforward to use a spacetime diagram (see Fig. (4)) to show that even though you and your friend agree that you both reached the finish line at the same time, you do not agree that you started the race at the same time. In the referee's frame, you and your friend started the race (denoted by event A and B, respectively) at the same time, and you and your friend finished at the same time (event C). Note that your and your friend's worldlines cross at C, so both you and your friend agree that you finished at the same time. However, your line of simultaneity going through event A intersects your friend's worldline at event B'. In other words, when you were just about to start the race, you observed that you friend had already started before you did. Similarly, your friend's line of simultaneity shows that when she was about to start the race, she observed that you had already started before her. Therefore you both think the other person cheated by starting the race early.

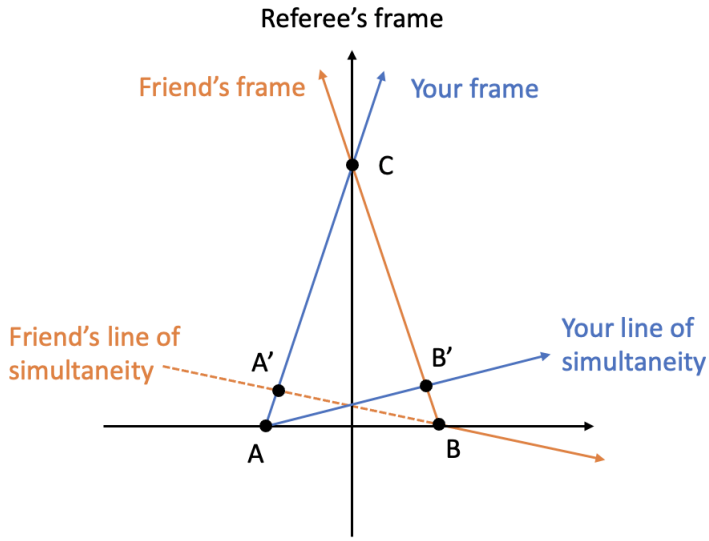

Figure 4: The spacetime diagram that illustrates problem 5. Again, the black, orange, and blue axes indicate the referee's, your friend's, and your frame, respectively, and the dashed blue and orange lines correspond to the line of simultaneity in your and your friend's frame, respectively. You and your friend started the race at event A and B, respectively, and both crossed the finish line at event C which indicates that you and your friend aggree that you both finished at the same time. However, the lines of simultaneity in your and your friend's frame show that both you and your friend think the other person started the race early.

6

First, let's consider the event at which the flash of light emitted from the origin reaches the end of the rod in frame R. We know that the rod is at rest in R, so the end of the rod is still located at $x = L$ when light reaches it, and it takes light a time of L to reach the end of the rod (recall that the speed of light is c in any inertial reference frame and we are setting $c = 1$, so $L/c = L$). The coordinates of the event in frame R is:

$$(t, x) = (L, L). \tag{13}$$

To find the coordinates of the same event (recall that an event itself is invariant of the reference frame but its coordinates are not) in frame R', we just need to perform the Lorentz transform on x and t. We find

$$x' = \gamma(v)(x + vt) = \sqrt{\frac{1+v}{1-v}}L \tag{14}$$

$$t' = \gamma(v)(t + vx) = \sqrt{\frac{1+v}{1-v}}L, \tag{15}$$

which are the coordinates of the event in frame R'.

7

a. The problem did not specify the origin of frame A (the Argonaut's frame) or frame B (the Barracuda's frame), so we can make the choice that is most convenient for us. The easiest choice for part a is to place frame A's origin at the head of the Argonaut and place frame B's origin at the tail of the Barracuda. The origins of the two frames coincide when the head of the Argonaut and tail of the Barracuda are aligned (call it event 1). In addition, we can set the clocks of the Argonaut and the Barracuda to both read time 0 at that instance. Then the coordinates of event 1 in the Argonaut's frame and in the Barracuda's frame are

$$(t_A^{(1)}, x_A^{(1)}) = (0, 0), \tag{16}$$

and

$$(t_B^{(1)}, t_B^{(1)}) = (0, 0), \tag{17}$$

respectively.

b. Let's call the Argonaut firing the torpedo event 2. In frame A, the torpedo is fired at the same time as event 1, so $t_A^{(2)} = t_A^{(1)} = 0$. We also know that the torpedo is fired from the Argonaut's tail, so $x_A^{(2)} = -L_0$. To find out the coordinates of event 2 in frame B, we just need to perform the Lorentz transform:

$$x_B^{(2)} = \gamma(v)(x_A^{(2)} + vt_A^{(2)}) = -\frac{L_0}{\sqrt{1-v^2}} \tag{18}$$

$$t_B^{(2)} = \gamma(v)(t_A^{(2)} + vx_A^{(2)}) = -\frac{v}{\sqrt{1-v^2}}L_0. \tag{19}$$

So, the coordinates of the torpedo being fired are $(t_A^{(2)}, x_A^{(2)}) = (0, -L_0)$ in frame A and $(t_B^{(2)}, x_B^{(2)}) = (-\frac{vL_0}{\sqrt{1-v^2}}, -\frac{L_0}{\sqrt{1-v^2}})$ in frame B.

c. Let's call the event at which the tail of the Argonaut and the head of the Barracuda are aligned event 3. To figure out the coordinates of event 3 in frame A, notice that event 3 happens at the tail of the Argonaut which is always at rest at $x_A = -L_0$ in frame A, so

$$x_A^{(3)} = -L_0.$$

To figure out the time of event 3 in frame A, we notice that the length of the Barracuda is contracted to be $L_0/\gamma(v)$ in frame A, so the Barracuda needs to travel a distance of $L_0 - L_0/\gamma(v)$ in order for its head to reach the tail of the Argonaut. That implies

$$t_A^{(3)} = \left(L_0 - \frac{L_0}{\gamma(v)}\right)/v = \left[1 - \frac{1}{\gamma(v)}\right]\frac{L_0}{v} = (1 - \sqrt{1-v^2})\frac{L_0}{v}. \tag{20}$$

Now we just need to perform the Lorentz transform to get the coordinates of event 3 in frame B:

$$x_B^{(3)} = \gamma(v)(x_A^{(3)} + vt_A^{(3)}) = -L_0 \tag{21}$$

$$t_B^{(3)} = \gamma(v)(t_A^{(3)} + vx_A^{(3)}) = \left(\frac{1}{\gamma(v)} - 1\right)\frac{L_0}{v} = -(1 - \sqrt{1-v^2})\frac{L_0}{v}. \tag{22}$$

To determine if the Barracuda gets hit or not, we just need to compare $t^{(2)}$ and $t^{(3)}$ in a particular frame. More specifically, if the torpedo is fired before the tail of the Argonaut and the head of the Barracuda are aligned, then the Barracuda is safe. If the Barracuda is safe in one reference frame, then it must be safe in all reference frames because whether the Barracuda gets hit or not should not depend on the observer. However, we will check in both frame A and frame B here to verify this.

First, it is clear that $t_A^{(3)} > t_A^{(2)}$ since $t_A^{(2)} = 0$ and $1 - \frac{1}{\gamma(v)} > 0$. So, according to the Argonaut, the Barracuda does not get hit. Now check if that is the case for the Barracuda, which will require a little more thought. To compare $t_B^{(3)}$ and $t_B^{(2)}$, we can do

$$\begin{aligned} t_B^{(3)} - t_B^{(2)} &= \frac{L_0}{v}\left(\frac{1}{\gamma(v)} - 1\right) + v\gamma(v)L_0 \\ &= \frac{L_0}{\gamma(v)v}(1 - \gamma(v) + v^2\gamma^2(v)) \\ &= \frac{L_0}{\gamma(v)v}\left(1 - \frac{1}{\sqrt{1-v^2}} + \frac{v^2}{1-v^2}\right) \\ &= \frac{L_0}{\gamma(v)v}\left(\frac{1}{1-v^2} - \frac{1}{\sqrt{1-v^2}}\right) \\ &= \frac{L_0}{\gamma(v)v}\frac{1}{\sqrt{1-v^2}}\left(\frac{1}{\sqrt{1-v^2}} - 1\right). \end{aligned} \tag{23}$$

Note that $\frac{1}{\sqrt{1-v^2}} - 1 > 0$ and so are the rest of the factors in the last line of Eq. (23), so we conclude that $t_B^{(3)} - t_B^{(2)} > 0$, or equivalently $t_B^{(3)} > t_B^{(2)}$. This means the Barracuda does not get hit in frame B either, just as expected.